

Correction formulas for the Mølmer-Sørensen gate under strong driving

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The Mølmer-Sørensen gate is a widely used entangling gate for ion platforms with inherent robustness to trap heating. The gate performance is limited by coherent errors, arising from the Lamb-Dicke (LD) approximation and sideband errors. Here, we provide explicit analytical formulas for errors up to fourth order in the LD parameter, by using the Magnus expansion to match numerical precision, and overcome significant, orders-of-magnitude underestimation of errors by previous theory methods. We show that fourth order Magnus expansion terms are unavoidable, being in fact leading order in LD, and are therefore critical to include for typical experimental fidelity ranges. We show how these errors can be dramatically reduced compared to previous theory by using analytical renormalization of the drive strength, by calibration of the Lamb-Dicke parameter, and by the use of smooth pulse shaping.

I. INTRODUCTION

A ubiquitous aspect of quantum processor design is the need to provide a strong contrast, or on/off ratio, between entanglement coupling vs. idling and local single-qubit gates. To this effect, the standard approach is to introduce new coupling elements, whose auxiliary Hilbert space provides a means through which entanglement can temporarily be mediated, for example via intermediate cavity bus [1–3], non-linear resonators [4–6], motional trapping modes [7–13], intermediary spins [14, 15], etc.

To fully leverage these coupling elements, and avoid the onset of decoherence, it is desirable to turn on the effective coupling between qubits as strongly as possible, for example through the use of strong driving fields. The prototypical example of this is the so-called Mølmer-Sørensen gate (MSG). This technique was first introduced in [16] to be resilient against trap heating within the Lamb-Dicke and weak driving perturbative approximations [17–23]. And yet, it is important to recognize that wide adoption of the gate largely occurred after a further advancement, whereby strong driving could be reinstituted at the cost of the collective ion motion being temporarily activated before returning to its initial state [24].

In the wider context of trapped ion quantum computing [25–27], the platform has blossomed to one of the leading candidates for housing quantum information, boasting long coherence times around 1 s to 50 s [28, 29], and infidelities for two-qubit gates on the order of 1×10^{-3} at a gate duration of 1.6 μ s to 300 μ s [8–11].

Further advancements hinge on proper theoretical understanding of the effect of strong driving on the operation with proper accounting of the excitation of the oscillator states, and potentially reduced gate durations. As with the oscillator-based couplings in other platforms, there are limited toolboxes available for tracking the dynamical effects of strong driving, the two most common

being the Dyson and Magnus expansions. In the Mølmer-Sørensen gate, the standard argument, dating back 25 years to Ref. [24], argues that, within the Lamb-Dicke approximation, an effective Hamiltonian in the Magnus expansion leads to only a single order being nonzero and accounting for the entanglement. The gate itself is second order in the Lamb-Dicke parameter η , while the error is fourth order in η , appearing in the same Magnus order, and so dictates the ultimate error scaling of the gate.

In this paper, we demonstrate that, on the contrary, there are multiple non-zero $\mathcal{O}(\eta^2)$ terms in the effective Hamiltonian that contribute even more significantly to the gate error. These errors are higher order in the Rabi frequency, but since this cannot be considered a small parameter (compared especially to the sideband detuning), these are fundamental to the functioning of the gate. We show that, in general, neglecting these terms leads to large detectable errors and deviations from experiment. These translate for typical parameters to circa 10 % Rabi frequency mismatch, and, without in-situ recalibration, 10-100 times larger error than theoretically predicted.

We provide analytical expressions for the leading orders of those terms and give a correction term for the drive strength Ω that can partially compensate for the error caused by those terms. Our model results in a bound on the gate infidelity of about 10^{-3} for typical parameters, which matches well with experimental fidelity measurement results [9, 10, 30–33]. We also motivate analytically and show numerically how it is – depending on the parameter regime – of advantage to use pulse shaping, to improve the gate fidelity closer to 10^{-4} . This can help to approach fidelities needed for fault-tolerant quantum computing [34, 35].

The remaining structure of the paper is as follows: In Section II, we (A) review the Magnus and the Dyson methods to calculate the propagator of the composite system; (B) discuss the effect of strong driving on the convergence the Magnus expansion; and (C) define the

system Hamiltonian and its expansion into Taylor- and Fourier coefficients. Next, in Section III, we explain a way to calculate arbitrary orders of the Magnus expansion for the given system, and investigate resonance conditions for turning on and off certain transitions. In Section IV, we provide analytical expressions for the different terms of the Magnus expansion, for the typical case of a rectangular drive amplitude, giving correction formulas accounting for higher Magnus order errors. We also evaluate numerically the importance of different system parameters on those errors. In Section V, we further provide analytical expressions and evaluate terms numerically, this time for shaped drive amplitudes. We finally summarize in Section VI.

II. MODEL AND METHODS

A. Time propagation

We first discuss the general methodology for obtaining the time propagation of our driven system. We want to first emphasize that, for a system such as the Mølmer-Sørensen gate, the drive dependence cannot in general be described with perturbation theory, since it may be larger than other relevant energy scales. Therefore, where multiple small parameters may exist, these different contributions should be properly accounted to avoid early truncation.

To this end, we describe the two most common expansions to obtain the propagator $\hat{U}(T)$ of a quantum system that gives the time evolution between time 0 and time T .

The Dyson expansion [36] gives the propagator as

$$\hat{U}(T) = \mathbb{1} + \sum_{k=1}^{\infty} \hat{P}_k(T), \quad (1)$$

where each term of the Dyson sum can be written as a time-ordered integral

$$\hat{P}_k(T) = \left(-\frac{i}{\hbar}\right)^k \int_0^T dt_1 \cdots \int_0^{t_{k-1}} dt_k \hat{H}(t_1) \cdots \hat{H}(t_k). \quad (2)$$

This expansion is mostly used for systems with a small perturbative interaction Hamiltonian.

An expansion with somewhat better convergence properties is the Magnus expansion [37]. It is exact for systems that are bosonic, and often converges quickly for fast oscillating systems, as is the case here. The Magnus propagator [38]

$$\hat{U}(T) = e^{-i \sum_{k=1}^{\infty} \hat{Z}_k(T)} \quad (3)$$

is calculated with a time-independent dimensionless effective Hamiltonian $\sum_{k=1}^{\infty} \hat{Z}_k(T)$, which gives after time T the same dynamics as the original Hamiltonian. The effective Hamiltonian is the sum of the different orders of

the Magnus expansion. The first three orders are

$$\begin{aligned} \hat{Z}_1 &= \frac{1}{\hbar} \int_0^T dt \hat{H}(t), \\ \hat{Z}_2 &= \frac{i}{2\hbar^2} \int_0^T dt \int_0^t d\tau [\hat{H}(t), \hat{H}(\tau)], \\ \hat{Z}_3 &= -\frac{1}{6\hbar^3} \int_0^T dt \int_0^t d\tau \int_0^\tau d\tau' \left([[\hat{H}(t), \hat{H}(\tau)], \hat{H}(\tau')] \right. \\ &\quad \left. + [\hat{H}(t), [\hat{H}(\tau), \hat{H}(\tau')]] \right). \end{aligned} \quad (4)$$

There are also formulas for higher orders [39], which we do not explicitly write down here due to their complexity.

Salzmann [40] gives instead a formula which expresses the orders of the Magnus expansion in terms of different orders of the Dyson expansion. Using his result, one can easily show that if $i\hat{P}_1 = \hat{Z}_1 = 0$, i.e., if the perturbation is unbiased, the relations between Magnus and Dyson expansion are

$$\begin{aligned} \hat{Z}_2 &= i\hat{P}_2, \\ \hat{Z}_3 &= i\hat{P}_3, \\ \hat{Z}_4 &= i \left(\hat{P}_4 - \frac{1}{2} \hat{P}_2^2 \right), \\ \hat{Z}_5 &= i \left(\hat{P}_5 - \frac{1}{2} (\hat{P}_2 \hat{P}_3 + \hat{P}_3 \hat{P}_2) \right). \end{aligned} \quad (5)$$

Clearly, Eq. (5) shows that the first two nonzero orders of the Magnus expansion equal the first nonzero orders of the Dyson expansion, except for a phase. Starting at fourth order, the terms for Dyson and Magnus expansion differ. This connection between Magnus and Dyson expansion is quite powerful, because, unlike the terms of the Magnus expansion, the terms of the Dyson expansion do not involve nested commutators. In the following sections, we calculate the Magnus expansion using Eq. (5), allowing us to efficiently and analytically track the role of different errors. Note that while the Dyson expansion gives an approximated propagator, the Magnus expansion gives an effective time-independent Hamiltonian, which is then used to calculate the propagator.

B. Perturbations to driven harmonic oscillator

To underscore the existence of error mechanisms arising from low-perturbation yet higher-order scattering processes under strong driving, we first show how they emerge under a generic driven Hamiltonian.

We consider a perturbative coupling between a coupler element (here harmonic oscillator) and another (computational) subsystem, and work in an appropriate rotating frame or interaction picture. This encompasses the Hamiltonian of the Mølmer-Sørensen gate, but also coupling mechanisms for a wide variety of other fast two-qubit gates [1–13] and readout [41–43] mechanisms.

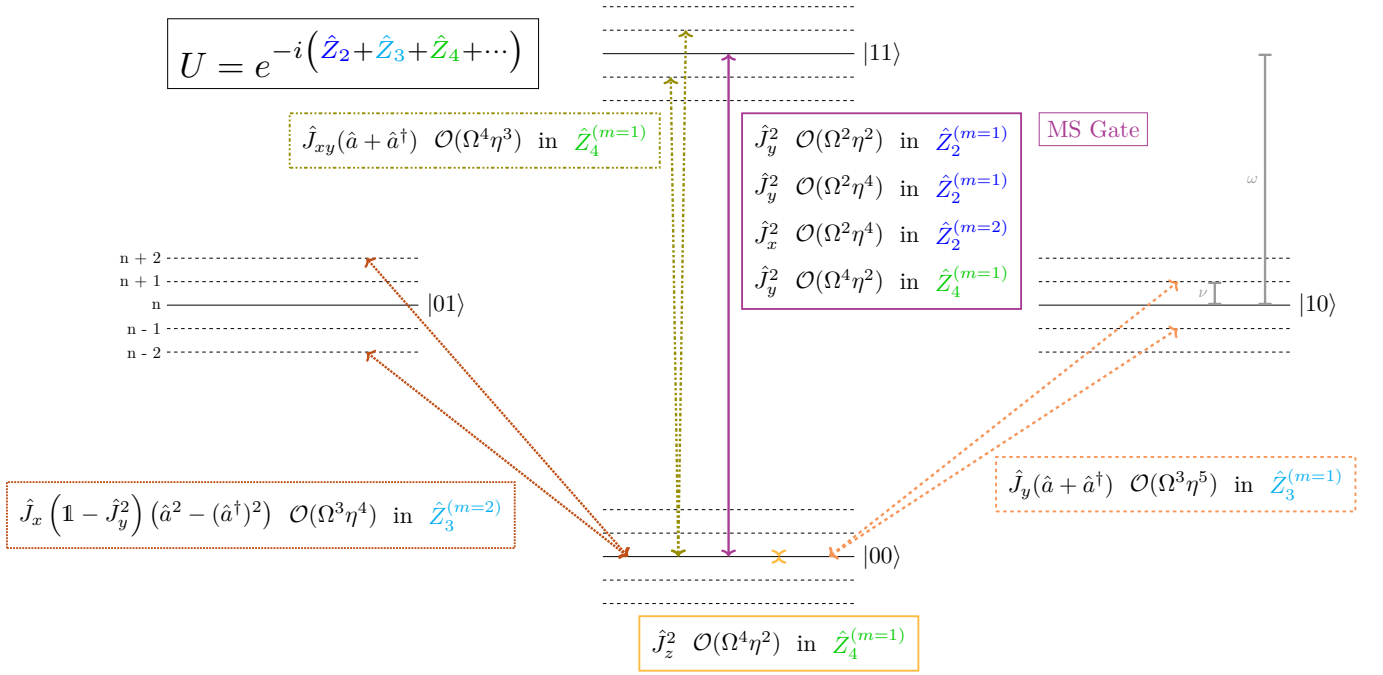


FIG. 1: Energy level diagram for the system in the interaction picture, with transitions connected to the $|00\rangle$ state being shown. The straight black lines depict the energy of the collective qubit state, and the dashed black lines depict coupled qubit and motional mode states. The solid pink arrow between $|00\rangle$ and $|11\rangle$ shows the transitions which create the entangling gate. Those errors can be corrected by adjusting the drive amplitude. The dashed-dotted green arrows left to the solid arrows show transitions that create an entangling operation in the qubit subspace, but additionally cause leakage into different motional mode states. The term proportional to $\hat{J}_z^2$ (yellow, at $|00\rangle$) only adds a phase. The dashed orange lines on the right-hand side denote first-sideband ($m = 1$) off-resonant error terms, and the dotted red lines on the left-hand side denote the second-sideband ($m = 2$) off-resonant error terms. Those terms can be minimized by time-shaping the drive amplitude. The full terms are listed in Table I. The lines denoting the transitions are just a sketch to give an intuition on the impact of each term, so not all potential transitions are shown.

We start with generically expanding the interaction in orders of a small parameter λ ,

$$\hat{H} = \hat{H}_0(t) + \lambda \hat{H}_1(t) + \mathcal{O}(\lambda^2), \quad (6)$$

which acts on the composite space $\mathcal{H}_{\text{HO}} \otimes \mathcal{H}_{\text{qubits}}$. Now, we consider the effect of different resonance offsets (ω_k) accompanying the different perturbation orders of the Hamiltonian ($\hat{H}_k$), or more precisely

$$\hat{H}_k(t) = \hbar \Omega(t) \hat{a}^k e^{-i\omega_k t} \otimes \hat{O}_k + H.c., \quad (7)$$

where Ω is the drive amplitude, which we cannot always assume is smaller than the resonance offsets ω_k due to strong driving, $\hat{O}_k$ are operators acting on the qubit subspace, and $\hat{a}, \hat{a}^\dagger$ are the annihilation and creation operators, respectively. Note that we have fixed the particular case of a common drive amplitude and specific powers of the harmonic oscillator operators, but this is simply done for concreteness and readability.

In the first order Magnus expansion, we have $\hat{Z}_1 = \frac{1}{\hbar} \int_0^T dt \hat{H}(t)$ corresponding to oscillating (off-resonant) terms. Often these are enforced to integrate to zero, but not strictly needed for our purposes in this subsection.

The second order of the Magnus expansion is given by

$$\hat{Z}_2 = \frac{i}{2\hbar^2} \int_0^T dt \int_0^t d\tau [\hat{H}(t), \hat{H}(\tau)], \quad (8)$$

where the commutator can be broken down as

$$\begin{aligned} [\hat{H}(t), \hat{H}(\tau)] &= [\hat{H}_0(t), \hat{H}_0(\tau)] + \lambda^2 [\hat{H}_1(t), \hat{H}_1(\tau)] \\ &\quad + \lambda [\hat{H}_1(t), \hat{H}_0(\tau)] + \lambda [\hat{H}_0(t), \hat{H}_1(\tau)] + \dots \end{aligned} \quad (9)$$

The important point to note is that the first line of Eq. (9) corresponds to resonant driving while the second does not, since the zeroth and first order frequency offsets are non-commensurate. In particular, in later sections the λ^2 term will correspond to the MSG interaction.

The third order of the Magnus expansion similarly contains a sum of commutators as in Eq. (4),

$$[[\hat{H}(t), \hat{H}(\tau)], \hat{H}(\tau')] + [\hat{H}(t), [\hat{H}(\tau), \hat{H}(\tau')]], \quad (10)$$

but which are always non-resonant, since there is an odd number of factors. Unlike in the first order, these are

generally not calibrated to integrate to zero, but these errors will still be relatively small.

Meanwhile, going to the fourth order $\hat{Z}_4$, the Magnus expansion will introduce commutators of the form

$$\lambda^2 \left[\left[\left[\hat{H}_0(t), \hat{H}_1(\tau) \right], \hat{H}_0(\tau') \right], \hat{H}_1(\tau'') \right], \quad (11)$$

and other permutations thereof, that is where we have an even number of H_0 and H_1 factors.

We see that we have a resonant coupling interaction of the same perturbation order as the one arising from $\hat{Z}_2$. Because we drive relatively strongly, the fact that the interaction is fourth order in Ω does not in general suppress the transition.

The general takeaways are twofold. The first is that the perturbation order of the Hamiltonian need not be linked with increasing expansion orders when the driving itself is non-perturbative. This is especially true in many-level systems where spurious resonances may arise. The second takeaway is that the expansion is non-monotonic, e.g., with the error increasing for the first four orders before eventually reducing again.

Therefore, in this general case, it is insufficient to terminate the series at low order. Fig. 1 shows the parameter scaling of errors in the Magnus expansion for the MSG Hamiltonian studied in this paper, where $\lambda = \eta$ is the Lamb-Dicke factor. Such higher-order processes must be carefully dealt with whenever coupler elements are driven strongly to accelerate entangling operations between subsystems.

C. MSG System Hamiltonian

We model the trapped ion system as two qubits coupled to a harmonic oscillator [44]. Despite the capability of the Mølmer-Sørensen gate to operate pairwise on ion chains of arbitrary length, we focus for the sake of simplicity in this paper on two-qubit gates, without loss of generality. The Hamiltonian of the uncoupled two-qubit system is $H_q = \hbar\omega_q \hat{J}_z$ with qubit transition frequency ω_q and a collective spin operator defined by

$$\hat{J}_i = \frac{1}{2} (\mathbb{1} \otimes \hat{\sigma}_i + \hat{\sigma}_i \otimes \mathbb{1}), \quad (12)$$

where σ_i (for $i = x, y, z$) are the Pauli operators.

Along the z-axis of the trap, the ions are confined in a harmonic effective potential. Their motion can be described in terms of motional modes [45]. When there are two ions, the two available modes are the so-called breathing mode and center-of-mass mode. In the following, we take only one of the modes into account, which is a valid assumption as long as the drive detuning is close to the frequency of the mode [24] (see Appendix C). In the Hamiltonian, the motional mode is modeled as a harmonic oscillator with frequency ν , and lowering and raising operator $\hat{a}$, $\hat{a}^\dagger$, respectively.

The interaction between the systems is invoked by a classical light field with drive envelope $\Omega(t)$ and wave number k along the trap x-axis [46]. The entangling operation of the Mølmer-Sørensen gate is generated by driving two laser frequencies at the same time [24]. When each laser is detuned by $\pm\delta$ from the qubit frequency ω_q , the Hamiltonian can be written as

$$\hat{H} = \hbar\omega_q \hat{J}_z + \hbar\nu \hat{a}^\dagger \hat{a} + 2\hbar\Omega(t) \sum_{\mu=\pm 1} \cos(k\hat{q} - (\omega_q + \mu\delta)t) \hat{J}_\mu, \quad (13)$$

with $\hat{q} = \sqrt{\frac{\hbar}{2M\nu}} (\hat{a} + \hat{a}^\dagger)$, where M is the mass of the ion, and $\mu = \pm 1$ is the sign of the detuning.

Moving to an interaction picture with respect to both qubits and the motional mode and dropping terms that rotate at twice the qubit frequency (rotating wave approximation), the Hamiltonian can be written as

$$\hat{H} = \hbar\Omega(t) \cos(\delta t) \left(\hat{J}_+ e^{ik\hat{q}(t)} + \hat{J}_- e^{-ik\hat{q}(t)} \right), \quad (14)$$

where $\hat{J}_\pm = (\hat{J}_x \pm i\hat{J}_y)$ are the collective spin raising and lowering operators.

In the following sections, we need to calculate integrals of products of the Hamiltonian at different times. Thus, it is useful to expand the Hamiltonian as a power series in the raising and lowering operators that act on the motional mode subspaces. To achieve this, we first express the operator for the motional mode in terms of the creation and annihilation operators ($\hat{a}^\dagger$, $\hat{a}$) as $k\hat{q}(t) = \eta (\hat{a}e^{-i\nu t} + \hat{a}^\dagger e^{i\nu t})$, where η is the Lamb-Dicke factor, which describes the strength of the coupling between the electronic state and the motional mode of the ion. Then, we use the Baker-Campbell-Hausdorff (BCH) formula [47] to factorize the operator exponentials as

$$e^{i\eta(\hat{a}e^{-i\nu t} + \hat{a}^\dagger e^{i\nu t})} = e^{i\eta\hat{a}^\dagger e^{i\nu t}} e^{i\eta\hat{a}e^{-i\nu t}} e^{\frac{\eta^2}{2}}. \quad (15)$$

Next, we expand both exponentials in Eq. (15) via Taylor series and group terms together with an index m for the sideband order (i.e. the number of phonons created or annihilated). Meanwhile, we Fourier-decompose the time-dependent parts in Eq. (14) and write the drive amplitude over the time interval $[0, T]$ as a Fourier series

$$\Omega(t) = \Omega \sum_{M \in \mathbb{Z}} c_M e^{i\frac{2\pi}{T} M t}, \quad (16)$$

with $c_{-M} = c_M^*$, and where in principle we truncate the series at a frequency for bandwidth considerations. Inserting into Eq. (14), this leads to

$$\hat{H}(t) = \hbar\Omega \sum_{M \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} \sum_{\mu=\pm 1} c_M e^{i(\frac{2\pi}{T} M + m\nu + \mu\delta)t} \hat{J}_\mu \hat{A}_m, \quad (17)$$

where operations on the qubit subspace are described by the collective spin operator,

$$\hat{J}_m = \frac{1}{2} \left(\hat{J}_+ + (-1)^m \hat{J}_- \right) = \begin{cases} \hat{J}_x & \text{if } m \text{ even} \\ i\hat{J}_y & \text{if } m \text{ odd} \end{cases}, \quad (18)$$

which acts on both qubits. The transition in the motional mode under absorption or emission of m phonons is given by the m -th sideband transition operator

$$\hat{A}_m = e^{-\frac{1}{2}\eta^2} \sum_{k=\max(0,-m)}^{\infty} \frac{\eta^{2k+m} i^{2k+m}}{(m+k)!k!} (\hat{a}^\dagger)^{k+m} (\hat{a})^k. \quad (19)$$

We next proceed to calculate the Magnus expansion for the Hamiltonian Eq. (17), order by order, so that we can see the effect of each term, and how some terms can be corrected.

III. RESONANCE CONDITIONS

Our goal is to set conditions to suppress certain transitions in Fig. 1, while enabling the required MSG qubit double-excitation transitions. For this purpose, we associate the resonance conditions with beat-note frequencies

$$N(m, \mu) = \underbrace{\frac{\nu T}{2\pi}}_K m + \underbrace{\frac{\delta T}{2\pi}}_L \mu, \quad (20)$$

with dimensionless gate duration K and dimensionless laser detuning frequency L . As before, m is the sideband order and μ is the detuning sign. Only the zeroth harmonic of the pulse is included in this section, i.e. $M = 0$, for better readability. The generalized versions of the formulas are provided in Appendix A.

In this notation, the Hamiltonian Eq. (17) takes the simplified form

$$\hat{H}(t) = \hbar\Omega \sum_{m=-\infty}^{\infty} \sum_{\mu=\pm 1} e^{i\frac{2\pi}{T}N(m,\mu)t} \hat{J}_m \hat{A}_m. \quad (21)$$

While the Magnus expansion can be very cumbersome beyond second order, we benefit here from the simplification that $Z_1 = 0$, which is required to remove single qubit driving, whose details are given below. In effect, this means that, using Eq. (5), calculation of the k -th order Magnus expansion terms can be reduced in its time dependence to solving nested integrals of the form

$$I_{N_1, \dots, N_k} = \int_0^T dt_k \int_0^{t_k} dt_{k-1} \dots \int_0^{t_2} dt_1 e^{i\frac{2\pi}{T} \sum_{j=1}^k N_j t_j}, \quad (22)$$

where the exponent consists of the sum of the exponents of Eq. (21) evaluated at different times t_j and for different combinations of the beat-note frequencies $N_j \equiv N(m_j, \mu_j)$. Finding appropriate solutions to the conditions will allow us then to significantly simplify the time-ordered operators in Eq. (2), rewritten here as

$$\hat{P}_k(t) = \left(-\frac{i}{\hbar}\right)^k \prod_{j=1}^k \left(\sum_{m_j=-\infty}^{\infty} \sum_{\mu_j=\pm 1} \hat{J}_{m_j} \hat{A}_{m_j} I_{N_j} \right), \quad (23)$$

which factors out the time dependence.

We proceed to calculate the leading orders of the nested integrals analytically and then verify their accuracy numerically. A main challenge, especially for fast oscillating numerics, is to ascertain whether for single coefficients N_j the integrals are zero, and whether they combine to sum up to zero, each corresponding to different resonance conditions.

For the evaluation of the higher order terms of the Magnus expansion in this paper, we thus used integration by parts to simplify the integrals. The analytical forms and their numerical evaluation can then be calculated using the SymPy [48] and NumPy [49] python packages, respectively.

A. First order condition

The resonance condition for the first Magnus order results from the integral I_{N_1} , which produces $\hat{Z}_1 = i\hat{P}_1$ given by

$$\hat{Z}_1 = \Omega \sum_{m=-\infty}^{\infty} \sum_{\mu=\pm 1} \underbrace{\frac{e^{i2\pi N(m,\mu)} - 1}{i(m\nu + \mu\delta)}}_{\text{resonance condition}} \hat{J}_m \hat{A}_m. \quad (24)$$

This term describes the phonon-assisted flipping of the qubits. For the entangling gate, we want to suppress single qubit flips at the final time T , so we tune the parameters such that $\hat{Z}_1 = 0$. This requires that the beat-note index must be a nonzero integer

$$N(m, \mu) \in \mathbb{Z}^*, \quad (25)$$

which implies that also K and L are integers. For constant drive amplitude $M = 0$, fulfilling Eq. (25) can be ensured with a laser detuning δ that is not a multiple of the trap frequency ν .

Note that due to strong driving (compared to the detuning), the motional mode will nonetheless be excited, and only at the final time does it revert to zero. If, on the other hand, the laser detuning is on resonance with the motional mode frequency, $\delta = \nu$, the sideband transition remains populated also at the final time [44]. Small detuning of the laser frequency will violate Eq. (25) and thus lead to errors in $\hat{Z}_1$ [50], but is outside the scope of the current work.

B. Second order condition

In the second order, we plug the Dyson term, Eq. (23), into the Magnus expansion, Eq. (3), and obtain

$$\hat{Z}_2 = \frac{i}{2} \Omega^2 \sum_{m_1, m_2=-\infty}^{\infty} \sum_{\mu_1, \mu_2=\pm 1} \hat{J}_{m_1} \hat{J}_{m_2} \hat{A}_{m_1} \hat{A}_{m_2} I_{N_1, N_2}, \quad (26)$$

with the resonance integral I_{N_1, N_2} given by Eq. (22).

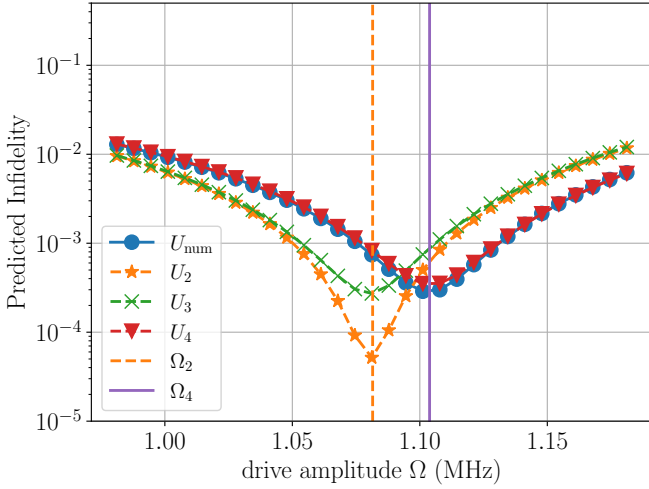


FIG. 2: Predicted average infidelity as a function of the drive amplitude Ω calculated with $\hat{U}_{\text{num}}$ and different Magnus expansion orders $\hat{U}_2, \hat{U}_3, \hat{U}_4$ in Eq. (42). The parameters are $\eta = 0.18$, $K - L = \frac{\nu - \delta}{2\pi} T = 3$, $K = 28$, $\nu/(2\pi) = 1$ MHz, and $\bar{n} = 2 \times 10^{-2}$. The theoretical prediction of Ω_2 (Eq. (35)) is depicted with the dashed orange vertical line. The theoretical prediction of the minimum Ω_4 (Eq. (40)) is shown with the solid purple vertical line. The Magnus orders $\hat{Z}_k$ are calculated taking into account the first three sideband orders $|m_i| \leq 3$.

We focus on the case where $N_j \in \mathbb{Z}, \forall j$. Then, the non-vanishing contributions to the integral are:

$$I_{N_1, N_2} = \begin{cases} \frac{T^2}{2} & N_1 = 0, N_2 = 0 \\ \frac{iT^2}{2\pi N_2} & N_1 = 0, N_2 \neq 0 \\ -\frac{iT^2}{2\pi N_1} & N_1 \neq 0, N_2 = 0 \\ -\frac{iT^2}{2\pi N_2} & N_1 \neq 0, N_2 \neq 0, N_1 + N_2 = 0. \end{cases} \quad (27)$$

Here, we have four resonance conditions. $N_j = 0$ are the resonance conditions for the unwanted single-photon transitions. The fourth condition, where $N_j \neq 0$ and $N_1 + N_2 = 0$, is the resonance condition for a two-photon transition. This is what creates the desired Mølmer-Sørensen gate entanglement.

The expression for the second order of the Magnus expansion, Eq. (26), can be further simplified if the single photon transitions, Eq. (24), are suppressed. Assuming that the detuning δ of the laser from the qubit frequency is of the order of the trap frequency ν , which is required anyway in order to excite a single motional mode only, then the two-photon resonance condition can be rewritten as

$$N_1 + N_2 = 0 \iff (m_1 = -m_2) \wedge (\mu_1 = -\mu_2). \quad (28)$$

In Appendix C, we discuss this matter in more detail. The simplification in Eq. (28) allows rewriting of Eq. (26)

as

$$\hat{Z}_2 = \sum_n \left(\hat{J}_x^2 d_x^{(n)} + \hat{J}_y^2 d_y^{(n)} \right) \otimes |n\rangle\langle n|, \quad (29)$$

with the standard form factors [24] that are the sum over even m for d_x and over odd m for d_y ,

$$d_{x,y}^{(n)} = \mp \frac{\Omega^2 T^2}{2\pi} e^{-\eta^2} \sum_{m \text{ even, odd}} \sum_{\mu=\pm 1} \frac{1}{mK+\mu L} (-\eta^2)^{|m|} \left(L_{\min(n, n-m)}^{|\mu|}(\eta^2) \right)^2 \frac{\min(n, n-m)!}{\max(n, n-m)!}, \quad (30)$$

where $L_a^{(b)}$ is the associated Laguerre polynomial [44].

Note that Eq. (29) is diagonal in the motional mode subspace. Therefore, as can be seen from Fig. 1, the terms in $\hat{Z}_2^{(m=1)}$ only perform flips (simultaneously) on the two qubits.

However, these conditions alone do not ensure that the Hamiltonian generates the correct gate. As we discuss next in Sec. IV, recalibrating the weights of these and other terms affects the rotation angle about the $\hat{J}_x^2$ and $\hat{J}_y^2$ operators.

C. Higher orders

When $\hat{Z}_1 = 0$, the third order of the Magnus expansion similarly simplifies to $\hat{Z}_3 = i\hat{P}_3$. The calculation of this term involves calculation of a triple integral. Assuming that we suppress single-photon transitions, $N_j \in \mathbb{Z}^* \forall j$, the resonance integral of third order is

$$I_{N_1, N_2, N_3} = \frac{T^3}{4\pi^2 N_3} \left(\frac{\delta_{N_1+N_2}}{N_2} + \frac{\delta_{N_2+N_3}}{N_1} \right). \quad (31)$$

This integral is zero unless $N_1 + N_2 = 0$ or $N_2 + N_3 = 0$, which are the resonance conditions for the transitions. Because of the complexity of Eq. (31), there is no compact form for $\hat{Z}_3$ beyond Eq. (23), as there was for $\hat{Z}_2$ in Eq. (29). The same applies for $\hat{Z}_4$ and $\hat{Z}_5$, but we will provide the leading analytical error terms for $\hat{Z}_3$ and $\hat{Z}_4$, for the typical case of a rectangular pulse next, in Section IV A. We then evaluate numerically the significance of those terms in Section IV B.

IV. ERROR TERMS FOR RECTANGULAR DRIVE PULSES

In this section, we present the leading order terms of the Magnus expansion and Lamb-Dicke parameter η for the Mølmer-Sørensen gate driven with rectangular pulses. We examine the impact and correction possibilities in terms of the average fidelity.

Error	Operator	Term	Term at Ω_{LD}	Term at Ω_4
Gate	$\hat{J}_y^2$	$-\frac{K\Omega^2 T^2 \eta^2}{\pi(K^2 - L^2)}$	$-\frac{\pi}{2}$	$-\frac{\sqrt{2K}\pi L\eta(s - \sqrt{s^2 - (K^2 - L^2)})}{(K^2 - L^2)}$
$\hat{Z}_2^{(m=1)}$	$\hat{J}_y^2$	$\frac{K\Omega^2 T^2 \eta^4 (2n+1)}{\pi(K^2 - L^2)}$	$\frac{\pi\eta^2(1+2n)}{2}$	$\frac{\sqrt{2K}\pi L\eta^3(2n+1)(s - \sqrt{s^2 - (K^2 - L^2)})}{(K^2 - L^2)}$
$\hat{Z}_2^{(m=2)}$	$\hat{J}_x^2$	$-\frac{K\Omega^2 T^2 \eta^4 (2n+1)}{\pi(4K^2 - L^2)}$	$-\frac{\pi\eta^2(K^2 - L^2)(2n+1)}{2(4K^2 - L^2)}$	$-\frac{\sqrt{2K}\pi L\eta^3(2n+1)(s - \sqrt{s^2 - (K^2 - L^2)})}{4K^2 - L^2}$
$\hat{Z}_3^{(m=1)}$	$\hat{J}_y(\hat{a} + \hat{a}^\dagger)$	$-\frac{2K^2\Omega^3 T^3 \eta^5}{\pi^2(K^2 - L^2)^2}$	$-\pi\eta^2\sqrt{\frac{K}{2(K^2 - L^2)}}$	$-\frac{2^{\frac{7}{4}}\pi K^{\frac{5}{4}}L^{\frac{3}{2}}\eta^{\frac{7}{2}}(s - \sqrt{-K^2 + L^2 + s^2})^{\frac{3}{2}}}{(K^2 - L^2)^2}$
$\hat{Z}_3^{(m=2)}$	$\hat{J}_x(1 - \hat{J}_y^2)(\hat{a}^2 - \hat{a}^{\dagger 2})$	$\frac{K^2\Omega^3 T^3 \eta^4}{\pi^2(4K^2 - L^2)(K^2 - L^2)}$	$\frac{\pi\eta}{2}\sqrt{\frac{K}{2(K^2 - L^2)(4K^2 - L^2)}}$	$-\frac{2^{\frac{3}{4}}\pi K^{\frac{5}{4}}L^{\frac{3}{2}}\eta^{\frac{5}{2}}(s - \sqrt{s^2 - (K^2 - L^2)})^{\frac{3}{2}}}{(4K^2 - L^2)(K^2 - L^2)}$
$\hat{Z}_4^{(m=1)}$	$\hat{J}_{xy}(\hat{a} + \hat{a}^\dagger)$	$\frac{K\Omega^4 T^4 \eta^3}{\pi^3(K^2 - 4L^2)(K^2 - L^2)}$	$\frac{\pi(K^2 - L^2)}{4K\eta(K^2 - 4L^2)}$	$\frac{2\pi L^2\eta(s - \sqrt{s^2 - (K^2 - L^2)})^2}{(K^2 - L^2)(K^2 - 4L^4)}$
	$\hat{J}_z^2$	$-\frac{K\Omega^4 T^4 \eta^2}{4\pi^3 L^2(K^2 - 4L^2)}$	$-\frac{\pi(K^2 - L^2)^2}{16KL^2\eta^2(K^2 - 4L^2)}$	$\frac{\pi(K^2 - L^2 - 2s^2 + 2s\sqrt{s^2 - (K^2 - L^2)})}{2(K^2 - 4L^2)}$
	$\hat{J}_y^2$	$\frac{K\Omega^4 T^4 \eta^2}{4\pi^3 L^2(K^2 - L^2)}$	$\frac{\pi(K^2 - L^2)}{16KL^2\eta^2}$	$-\frac{\pi}{2} + \frac{\pi s(s - \sqrt{s^2 - (K^2 - L^2)})}{(K^2 - L^2)}$

TABLE I: Error terms for rectangular pulses. The second column shows the operator part of the term. The 3rd to 5th column show the coefficient of the operator. The 4th and 5th column are the same as the 3rd, except that $\Omega = \Omega_{\text{LD}}$ (Eq. (32)), and $\Omega = \Omega_4$ (Eq. (41) with $s = \sqrt{2K}L\eta(1 - \eta^2)$), respectively.

A. Analytical Expressions

The analytical expressions for the error terms are calculated by solving the time-ordered Dyson integrals Eq. (23) for small sideband numbers m . Moreover, when neglecting terms of the Magnus expansion that are off-diagonal in the motional mode subspace, we can also derive analytical expressions for the fidelity, as explained in detail in Appendix B.

When considering only terms in the Hamiltonian which are proportional to $\hat{J}_y^2$, a simple expression for the average fidelity can be obtained, see Eq. (B7). By employing the Bell fidelity, a simple expression can be derived for the case that terms proportional to $\hat{J}_x^2$ and $\hat{J}_z$ are taken into account as well, see Eq. (B3). Both expressions yield simple equations that can be solved for Ω to determine the value that optimizes the fidelity.

1. Errors and optimal drive amplitude at $\hat{Z}_2$

To the first nonzero order of η , taking $k = 0$ and the first sideband order $|m| = 1$ in Eq. (19), the form factors in Eq. (29) simplify, such that $d_x^{(0)} = 0$ and the second order Magnus term is $\hat{Z}_2 = d_y^{(0)}\hat{J}_y^2 = -\frac{\Omega^2 T^2}{\pi}\frac{K}{(K^2 - L^2)}\eta^2\hat{J}_y^2$.

The optimal parameters that fulfill a $\pi/2$ rotation around $\hat{J}_y^2$ can straightforwardly be solved (c.f. Appendix B), leaving us with

$$\Omega_{\text{LD}} = \frac{\pi}{T\eta}\frac{\sqrt{(K^2 - L^2)}}{\sqrt{2K}}, \quad (32)$$

where we assume that the motional mode is initially in the vacuum state $\bar{n} = 0$.

Taking the next order of the Lamb-Dicke expansion into account gives the Lamb-Dicke error term

$$\hat{Z}_2^{(m=1)} = \frac{K\Omega^2 T^2 \eta^4 (2n+1)}{\pi(K^2 - L^2)}\hat{J}_y^2 = \hat{J}_y^2 \mathcal{O}(\Omega^2 \eta^4), \quad (33)$$

which corresponds to the 2nd row in Table I. Taking also the next sideband order into account, with $|m| = 2$ in Section III B, gives the sideband error term (3rd row in Table I)

$$\hat{Z}_2^{(m=2)} = -\frac{K\Omega^2 T^2 \eta^4}{\pi(4K^2 - L^2)}(2n+1)\hat{J}_x^2 = \hat{J}_x^2 \mathcal{O}(\Omega^2 \eta^4). \quad (34)$$

Both errors have already been studied in the original work, Ref. [24], and can be minimized by further adjusting the drive amplitude

$$\Omega_2 = \frac{\pi}{T\eta}\sqrt{\frac{(K^2 - L^2)(4K^2 - L^2)}{2K(\eta^2(2L^2 - 5K^2) + (4K^2 - L^2))}}, \quad (35)$$

which can be found by inserting the form factors in Eq. (29) for $n = 0$ into the condition for the optimal fidelity (see Appendix B for more details).

However, these errors are only optimal in the approximation of zero motional mode population $\bar{n} = 0$, and

discounting the next orders of the Lamb-Dicke and Magnus expansion. We now discuss these latter corrections.

2. Errors at $\hat{Z}_3$

Now let us take a look at the higher orders of the Magnus expansion. In this subsection and the following, we discuss the error operators and the order of the terms in Ω and η , while full expressions are shown in Table I for better readability.

The first-sideband term of $\hat{Z}_3$ (4th row in Table I) for a rectangular pulse takes the form

$$\hat{Z}_3^{(m=1)} = \hat{J}_y(\hat{a} + \hat{a}^\dagger)\mathcal{O}(\Omega^3\eta^5), \quad (36)$$

which gives $\mathcal{O}(\eta^2)$ at $\Omega = \Omega_{\text{LD}}$. This term creates a phonon-assisted single-qubit flip. The second-sideband term for $\hat{Z}_3$ (5th row in Table I) is of order

$$\hat{Z}_3^{(m=2)} = \hat{J}_x \left(\mathbb{1} - \hat{J}_y^2 \right) (\hat{a}^2 - (\hat{a}^\dagger)^2) \mathcal{O}(\Omega^3\eta^4). \quad (37)$$

which gives $\mathcal{O}(\eta)$ at $\Omega = \Omega_{\text{LD}}$. This term causes squeezing [12], and population leakage into the motional mode. The energy level diagram in Fig. 1 depicts how the processes in $\hat{Z}_3$ cause population leakage in both the motional mode and qubit subspaces.

3. Errors and optimal drive amplitude at $\hat{Z}_4$

The fourth order of the Magnus expansion consists in the sum of three entangling operations

$$\begin{aligned} \hat{Z}_4^{(m=1)} = & \hat{J}_{xy} (\hat{a} + \hat{a}^\dagger) \mathcal{O}(\Omega^4\eta^3) \\ & + \hat{J}_z^2 \mathcal{O}(\Omega^4\eta^2) \\ & + \hat{J}_y^2 \mathcal{O}(\Omega^4\eta^2), \end{aligned} \quad (38)$$

where

$$\hat{J}_{\alpha\beta} = \frac{1}{2} (\hat{\sigma}_\alpha \otimes \hat{\sigma}_\beta + \hat{\sigma}_\beta \otimes \hat{\sigma}_\alpha).$$

The full expressions are listed in rows 6–8 of Table I. As sketched in Fig. 1, the first term in $\hat{Z}_4$, $\hat{J}_{xy} (\hat{a} + \hat{a}^\dagger)$, creates entanglement, but also causes population leakage in the motional mode. The second term, $\hat{J}_z^2$, only adds a phase to the target state. The third term, $\hat{J}_y^2$, contributes to the entangling gate. The leading terms of the first sideband order terms $\hat{Z}_2^{(m=1)} + \hat{Z}_3^{(m=1)} + \hat{Z}_4^{(m=1)}$ are proportional to $\hat{J}_y^2$. Inserting the resulting prefactor

$$d_y^{(n)} = -\Omega^2 T^2 \frac{K\eta^2(\eta^2 - 1)}{\pi(K^2 - L^2)} + \Omega^4 T^4 \frac{K\eta^2}{4\pi^3 L^2 (K^2 - L^2)} \quad (39)$$

into the condition for the Bell fidelity Eq. (B3) or for the average fidelity Eq. (B7) and setting $\mathcal{F} = 1$ yields a quadratic equation in Ω^2 ,

$$-\frac{K\Omega^2 T^2 \eta^2 (4\pi^2 L^2 \eta^2 - 4\pi^2 L^2 + \Omega^2 T^2)}{4\pi^3 L^2 (K^2 - L^2)} = \frac{\pi}{2}, \quad (40)$$

whose solution is the corrected optimal drive amplitude

$$\Omega_4^2 = \frac{\sqrt{2}\pi^2 L \left(s - \sqrt{s^2 - (K^2 - L^2)} \right)}{\sqrt{K} T^2 \eta}, \quad (41)$$

with $s = \sqrt{2K}L\eta(1 - \eta^2)$.

Since Ω_4 is $\mathcal{O}(\eta^{-\frac{1}{2}})$, in contrast to Ω_{LD} and Ω_2 which are $\mathcal{O}(\eta^{-1})$, the Lamb-Dicke orders of the terms of the Magnus expansion change when driving at Ω_4 instead of Ω_2 or Ω_{LD} , which is reflected in the last two columns of Table I.

B. Numerical Comparison

In the previous section, the analytical expressions for the leading error terms in the Magnus expansion were presented, summarized in Table I. This section focuses on numerically analyzing the impact of these error terms on the average fidelity (Eq. (B6)) of the Mølmer-Sørensen gate. We compare the infidelities $1 - \mathcal{F}_{\text{av}}$ calculated with different propagators. The fidelity representing the effect of the n -th order of the Magnus expansion is calculated with

$$\hat{U}_n := \exp \left\{ -\frac{i}{\hbar} \sum_{k=2}^n \hat{Z}_k \right\}. \quad (42)$$

The first order $\hat{Z}_1$ is always suppressed. The numerical expression for the Magnus terms $\hat{Z}_n$ are calculated using the formulas shown in the previous sections.

The operators are implemented in matrix form, with the operators acting on the motional mode subspace being truncated at $N_{\text{dim}} = 8$. The resonance integrals are calculated recursively using computer tools as described in Section III.

In the previous section, the analysis of the Magnus expansion orders was limited to the second to fourth order. To ensure the adequacy of this restriction, we calculated the infidelity including the fifth-order Magnus term as well, represented as $\hat{U}_5$. However, since the results show that the curves obtained with $\hat{U}_5$ nearly completely overlap with those calculated with $\hat{U}_4$, we have decided to not include them into the figures in order to avoid visual clutter.

To evaluate the accuracy of a truncated propagator in representing the actual system evolution, we estimate the propagator for infinite Magnus orders $\hat{U}_\infty$. This estimation is accomplished through numerical calculation of the propagator using the Trotter expansion [51, 52] which decomposes the propagator into products of exponential operators, each corresponding to a small time step. We define the numerical propagator as

$$\hat{U}_{\text{num}} := \prod_{n=0}^{N_t} e^{-\frac{i}{\hbar} H(n\Delta t)\Delta t}. \quad (43)$$

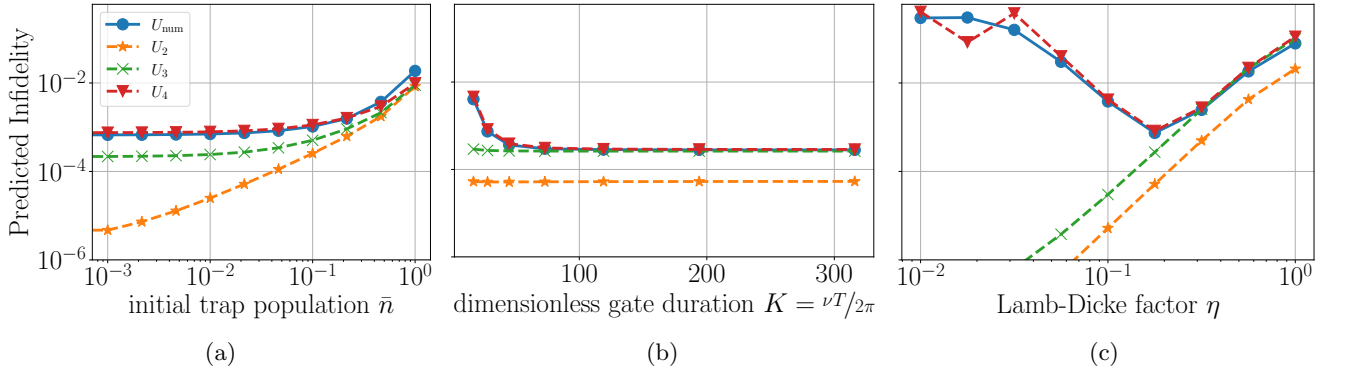


FIG. 3: Predicted average infidelity as a function of the population of the motional mode $\bar{n}$ (Fig. 3a), the dimensionless gate duration K (Fig. 3b), and the coupling strength η (Fig. 3c), respectively. The parameters are (if they are not varied): $\eta = 0.18$, $K - L = \frac{\nu - \delta}{2\pi} T = 3$, $K = 28$, $\nu / (2\pi) = 1$ MHz, $\bar{n} = 2 \times 10^{-2}$, and $\Omega = \Omega_2$ (Eq. (35)). U_n and U_{num} are the predicted infidelities calculated with Eq. (42) and Eq. (43), respectively.

To ensure that the time steps are sufficiently small, they are set to $\Delta t = \frac{1}{10f_{\text{max}}}$ where $f_{\text{max}} = \max\{M + mK + L\} \frac{2\pi}{T}$ is the largest frequency appearing in the Hamiltonian, Eq. (21). As the Magnus expansion terms for the numerical calculations are truncated after the third sideband order, we set $m = 3$ here.

The figures Figs. 2 to 4 were created using the following typical parameters [9, 35, 53, 54]: The Lamb-Dicke parameter is set to $\eta = 0.18$. The average trap population is chosen as $\bar{n} = 2 \times 10^{-2}$. For rectangular pulses, the dimensionless gate durations is $K = 28$, which corresponds to a gate duration of 28 μs for a trap frequency of $\nu / (2\pi) = 1$ MHz. The dimensionless laser detuning difference from the trap frequency $K - L$ is set to 3, corresponding to a detuning of $(\nu - \delta) / (2\pi) = 107$ kHz, illustrating that the smallest detuning of $K - L = 1$ may not always be optimal. Because the difference in fidelity for different values of $K - L$ is not significant and its behavior depends on the choice of other parameters, this article does not delve into the specifics of how the choice of $K - L$ affects the fidelity.

In Fig. 2, the predicted infidelities for the propagators $\hat{U}_n$ and $\hat{U}_{\text{num}}$ are displayed as a function of the drive amplitude Ω . The minima of the predicted infidelities calculated with $\hat{U}_2$ and $\hat{U}_3$, respectively, are approximately located at the drive amplitude Ω_2 , whose value was calculated using Eq. (35). The curves for $\hat{U}_4$ and $\hat{U}_{\text{num}}$ are nearly indistinguishable at off-resonance and are closely aligned around their minimum, which is approximately at the expected optimal Rabi frequency Ω_4 (Eq. (40)). The predicted infidelity of $\hat{U}_3$ at Ω_2 is approximately equal to the infidelities of $\hat{U}_4$ and $\hat{U}_{\text{num}}$ at Ω_4 . For the sake of consistency, in Figs. 3 and 4 we compare $\hat{U}_n$ to $\hat{U}_{\text{num}}$ with the drive amplitude set to $\Omega = \Omega_2$, but it is worth noting that the results are similar for $\Omega = \Omega_4$.

In the following, we provide a comparison between the infidelity calculated using different propagators as a function of various parameters, as shown in Fig. 3. As a general observation, in Fig. 3, it can be seen that there is a

strong agreement between the curve for $\hat{U}_4$ and the curve for $\hat{U}_{\text{num}}$. The dependence of the infidelity on the initial average motional mode population is plotted in Fig. 3a. At low temperatures, the curves calculated with $\hat{U}_2$ and $\hat{U}_3$ predict lower infidelity than those calculated with $\hat{U}_4$ and $\hat{U}_{\text{num}}$, highlighting their lack of predictive power. However, at higher temperature, all predicted infidelities increase monotonically and nearly converge. The infidelity is almost linear in the motional mode population $\bar{n}$, as expected from the terms which are linear in n in Table I.

Figure 3b illustrates the predicted infidelity as a function of the dimensionless gate time $K = \frac{\nu T}{2\pi}$, indicating a threshold around $K = 100$ below which the predicted infidelity increases significantly as K decreases. For gate times longer than $K \approx 100$, the curves representing $\hat{U}_3$ and $\hat{U}_4$ converge, whereas for gate times shorter than $K \approx 100$, the predicted infidelities diverge, as expected when the coherent errors increase and the ratio of Ω to the other frequencies in the system is increased. In particular, the predicted infidelities calculated with $\hat{U}_4$ and $\hat{U}_{\text{num}}$ increase significantly as K decreases. Meanwhile, the infidelities calculated with the lower orders of the Magnus expansion $\hat{U}_2$ and $\hat{U}_3$ remain nearly horizontal, indicating an inability to capture the full dynamics at short times, but the curve calculated with $\hat{U}_2$ lies an order of magnitude below the infidelity calculated with $\hat{U}_3$.

Figure 3c depicts the predicted infidelity as a function of the Lamb-Dicke parameter η . The curve calculated with $\hat{U}_2$ lies approximately an order of magnitude lower than the curve calculated with $\hat{U}_3$. The curves for $\hat{U}_4$ and $\hat{U}_{\text{num}}$ do approach the curve for $\hat{U}_3$ around $\eta = 0.3$ but this is in the higher error regime; on the other hand, they diverge significantly at smaller values of η , once again disagreeing strongly with the standard predictions. Notably, the curves calculated with $\hat{U}_2$ and $\hat{U}_3$ increase monotonically with η , while the curves calculated with $\hat{U}_4$ and $\hat{U}_{\text{num}}$ have a minimum around $\eta = 0.2$. This indicates that making the Lamb-Dicke parameter as small

as possible does not improve the infidelity of the Mølmer-Sørensen gate. Instead, there is an optimal value for the Lamb-Dicke parameter for the Mølmer-Sørensen gate.

C. Discussion

In Fig. 1, the terms of Section IV are visually represented. Through our analysis in the previous sections, it has become clear that the second order of the Magnus expansion, $\hat{Z}_2$, is not the only term that significantly impacts the gate fidelity. In fact, terms up to and including the fourth order $\hat{Z}_4$ contribute significantly to the fidelity. Although higher orders in the Magnus expansion may in principle be relevant as well, Section IV B demonstrates that they are typically not necessary for commonly used parameters.

In a model that accounts only for the second order of the Magnus expansion, one might be tempted to choose a small value for the Lamb-Dicke parameter η to minimize errors related to the Lamb-Dicke expansion. However, as demonstrated in Table I, reducing η decreases the error in $\hat{Z}_2$ and $\hat{Z}_3$, but also increases the error term in $\hat{Z}_4$. This indicates the existence of a sweet spot for η , as shown in Fig. 3c. The fourth order term $\hat{Z}_4$ leads, due to the additional term proportional to $\hat{J}_y^2$, to a shift in the optimal drive amplitude without significantly impacting the fidelity compared to $\hat{Z}_3$. Meanwhile, the contribution of $\hat{Z}_3$, compared to $\hat{Z}_2$, increases the infidelity, as illustrated in Table I and Fig. 2.

The temperature dependence of the error terms, as presented in Table I, reveals that certain terms are proportional to $\bar{n}$, consistent with the trend observed in Fig. 3a.

According to [24], one can neglect $\hat{Z}_3$, because the error in $\hat{Z}_2$ is $\mathcal{O}(n\eta^2)$, while the error in $\hat{Z}_3$ is $\mathcal{O}(\eta^2)$. They argue that deviations of the Lamb-Dicke approximation are typically caused by large values of n rather than η , making $\hat{Z}_3$ insignificant compared to $\hat{Z}_2$. Nevertheless, this perspective fails to consider that error terms related to higher sideband orders could have lower orders in the Lamb-Dicke expansion, as seen in the case of the second sideband order of $\hat{Z}_3$ (Eq. (37)). Moreover, due to improved cooling methods [53] since then, it is increasingly crucial to have expressions that remain valid at low temperatures.

V. PULSE SHAPING

So far, we have scrutinized the problem for a rectangular drive amplitude. However, the analysis in Section III is valid for arbitrary finite-window, smooth pulse shapes, with small modifications as given in Appendix A. Amplitude modulation experiments on the Mølmer-Sørensen gate have already been performed [35, 55], but to the authors' knowledge, no theoretical investigation has been realized so far. A straightforward candidate for a smooth pulse is a Gaussian pulse, but since it is not finite in time,

we fall back on a similar, and well-behaved pulse shape of the form

$$\Omega(t) = \Omega \sin^2\left(\frac{\pi}{T}t\right). \quad (44)$$

The $\sin^2$ pulse shape has also a narrower bandwidth (i.e. smaller maximal value of M), when compared to a truncated Gaussian pulse. This is potentially preferable since fewer unwanted transitions may be driven, and in general more compatible with keeping track of all the resonance conditions, discussed in Section III. Since the analytical results for a $\sin^2$ -pulse are lengthy, we only sketch some terms in this section and give further details in Appendix A and Appendix D.

A. Analytic expressions

The 2nd order of the Magnus expansion for a $\sin^2$ -pulse is of the form

$$\hat{Z}_{2,\sin^2} = \frac{K\Omega^2 T^2 \eta^2}{\pi} \left(\frac{p_y}{q_y} (1 - (2n+1)\eta^2) \hat{J}_y^2 + \frac{p_x}{q_x} (2n+1) \eta^2 \hat{J}_x^2 \right), \quad (45)$$

where $p_{x,y}$ and $q_{x,y}$ are polynomials in K and L , defined in Appendix D. From this expression, one can derive an optimal drive amplitude

$$\Omega_{\text{LD},\sin^2} = \frac{\pi}{\eta T \sqrt{2K}} \sqrt{\frac{q_y}{p_y}}. \quad (46)$$

The 3rd order of the Magnus expansion is of the form

$$\hat{Z}_{3,\sin^2} = \frac{K^2 \Omega^3 T^3 \eta^5}{\pi^2} \frac{p_3}{q_3} \hat{J}_y (\hat{a} + \hat{a}^\dagger), \quad (47)$$

where p_3 and q_3 are again polynomials in K and L .

One can show that the resulting error terms at the optimal drive amplitude are smaller than the error terms for a rectangular pulse, which have the same form, but different prefactors. This is, because the prefactor $\frac{p_3}{q_3}$ is smaller in the case of a $\sin^2$ -pulse than the respective one for a rectangular pulse given in Table I, for most realistic choices of K and L .

B. Numerical Comparison

We now compare the average infidelity, $1 - \mathcal{F}$, using a rectangular pulse $\Omega(t) = \Omega$ to the infidelity using a $\sin^2$ -pulse $\Omega(t) = \Omega \sin^2(\frac{\pi}{T}t)$. Explicitly, we compare $\hat{U}_n$ for $n = 2, \dots, 4$, to $\hat{U}_{\text{num}}$ using a $\sin^2$ -pulse, while also comparing to $\hat{U}_{\text{num}}$ using a rectangular pulse. The propagators $\hat{U}_n$, $\hat{U}_{\text{num}}$ are calculated with Eqs. (42) and (43), respectively.

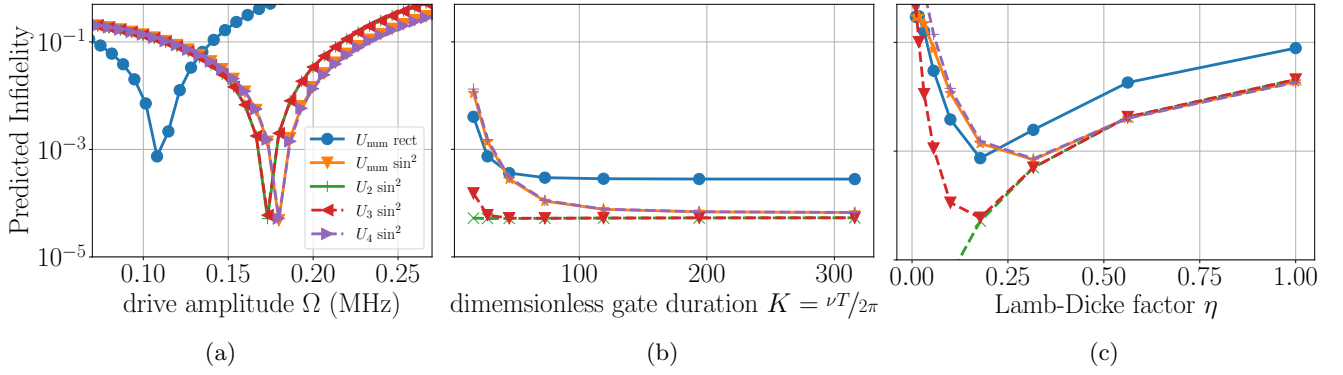


FIG. 4: Predicted average infidelity of the Mølmer-Sørensen gate $\sin^2$ -pulses and for comparison the numerical predicted infidelity for a rectangular pulse as a function of drive amplitude Ω , dimensionless gate duration K and Lamb-Dicke factor η , respectively. The parameters are (if they are not varied): $\eta = 0.18$, $K - L = \frac{\nu - \delta}{2\pi} T = 3$, $K = 28$, $\nu/(2\pi) = 0.1$ MHz, $\bar{n} = 2 \times 10^{-2}$. The data for Figs. 4b and 4c is calculated using $\Omega = \Omega_{2,\text{rect}}$, Eq. (35), for the rectangular pulse shape and $\Omega = 0.173$ MHz for the $\sin^2$ -pulse shape.

Comparing the predicted infidelities calculated with $\hat{U}_{\text{num}}$ as a function of Ω for a rectangular and a $\sin^2$ -pulse, one can see in Fig. 4a that the optimal (highest reached) drive amplitude differs for different pulse shapes, but that the minimal predicted infidelity for a $\sin^2$ -pulse is smaller than for a rectangular pulse. Because the optimal drive amplitudes differ, in Figs. 4b and 4c the fidelities are plotted at their respective minimal drive amplitude Ω_2 .

In Fig. 4b, the infidelity is shown as a function of the dimensionless gate duration K . In the low infidelity regime, the shaped pulses outperform the square pulses considerably, with both the full numerics and the fourth Magnus order closely agreeing. The graph also shows that inclusion of orders higher than three in the Magnus expansion (described by $\hat{U}_4$) results in an increase of the predicted infidelity as the value of K decreases, but only below a certain threshold that is dependent on η . At dimensionless gate times longer than $K \approx 50$, the calculation using a $\sin^2$ -pulse predicts a lower infidelity, but at gate times shorter than $K \approx 50$, the rectangular pulse yields a lower infidelity. One can see that for the $\sin^2$ -pulse, except for short gate times, there is a close agreement between the curve calculated with $\hat{U}_2$ and with $\hat{U}_3$, which implies that pulse shaping can probably compensate for the off-diagonal errors in $\hat{Z}_3$, unlike in Fig. 3b.

For the Lamb-Dicke parameter η , we see once again that the contributions of the higher orders in the Magnus expansion lead correctly to the existence of an optimal value for η , while instead the terms $\hat{Z}_2$ and $\hat{Z}_3$ decrease with decreasing η . The data in Fig. 4c is not detailed enough to tell which pulse shape in general leads to a smaller minimal predicted infidelity. For Lamb-Dicke parameters larger than the minimal value, the $\sin^2$ -pulse yields a much lower infidelity, but for small η , the rectangular pulse is slightly better. The behavior of the infidelity as a function of η is similar for rectangular pulses and $\sin^2$ -pulses.

C. Discussion

Figure 4 illustrates the potential for using shaped pulses to further reduce the error in the Mølmer-Sørensen gate. At a quantitative level, we see that while in the previous plots parameter searches limited infidelities to the 10^{-3} range, there are promising parameter regimes with $\sin^2$ -pulses that lead to errors below 10^{-4} . This can be explained mathematically, through the Magnus expansion, as arising out of generally smaller prefactors in front of the same error operators terms.

At a qualitative level, the behavior of the infidelity for varying parameters for a $\sin^2$ -pulse is similar to the behavior of the infidelity for a rectangular pulse. Moreover, for certain parameter regimes, the rectangular pulses actually outperform the $\sin^2$ -strategy. This is consistent both with conventional wisdom on pulse shaping and the nuances of the Mølmer-Sørensen gate. Indeed, square pulses are known to allow spectral hole burning through their characteristic sinc spectrum, while smooth pulses offer the possibility of reducing bandwidth overall and thereby avoid entire ranges of off-resonant errors. In the case of the Mølmer-Sørensen gate, the spectral conditions on beat-note indices not only reduce errors at low Magnus orders, but in certain cases also eliminate certain other terms, and so for a fortuitous choice of parameters it may indeed offer the best strategy. Smooth shaping allows nonetheless generally a safer option, and indeed seem to perform optimally in the best case. Lastly, the latter can be combined more easily with other control theoretic methods to potentially reduce errors even further.

VI. SUMMARY

We showed that calculating the Magnus expansion only up to the second order $\hat{Z}_2$ is not sufficient to adequately describe the Mølmer-Sørensen gate.

While the Lamb-Dicke expansion is indeed a valid approximation for the first nonvanishing Magnus order $\hat{Z}_2$, we show that higher orders of the Magnus expansion, and explicitly $\hat{Z}_3$ and $\hat{Z}_4$, cannot be neglected in general. The contribution of the third order $\hat{Z}_3$ increases significantly the infidelity, being of the same order $\mathcal{O}(\eta^4)$ in η as the Lamb-Dicke error terms in $\hat{Z}_2$. Moreover, the fourth order $\hat{Z}_4$ is even of the same order $\mathcal{O}(\eta^2)$ in η as the gate term in $\hat{Z}_2$, leading to a significant shift in the optimal drive amplitude. Due to being of different order in Ω , a sweet spot for the Lamb-Dicke coefficient η also emerges.

We give predictive expressions for the optimal drive amplitude (up to experimental calibration), and show qualitatively that an optimal η may be found, e.g. through numerical calibration. Due to the contribution of

the third order in the Magnus expansion $\hat{Z}_3$, the Mølmer-Sørensen gate fidelity is inherently limited, but with analytical expressions for leading terms of the first four Magnus terms, this opens potential avenues to suppress certain terms. For example, we show that an amplitude with envelope $\sin^2(\frac{\pi}{T}t)$ decreases the contribution of the third order of the Magnus expansion. The presented framework also opens new paths for application of optimal control methods on the Mølmer-Sørensen gate, such as the use of DRAG pulses [56]. It also makes it straightforward to investigate the effect of multichromatic beams.

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Appendix A: General expressions for the resonance integral with shaped pulses

1. First order

The general expression for the first order of the Magnus expansion Eq. (24) is

$$\hat{Z}_1 = \Omega \sum_{M,m=-\infty}^{\infty} \sum_{\mu=\pm 1} c_M \underbrace{\frac{e^{i(\frac{2\pi}{T}M+m\nu+\mu\delta)T} - 1}{i(\frac{2\pi}{T}M+m\nu+\mu\delta)}}_{\text{resonance condition}} \hat{J}_m \hat{A}_m. \quad (\text{A1})$$

Using the beat-note index

$$N(M, m, \mu) := M + \underbrace{\frac{\nu T}{2\pi} m}_K + \underbrace{\frac{\delta T}{2\pi} \mu}_L \quad (\text{A2})$$

the Hamiltonian can be written more compactly as

$$\hat{H}(t) = \hbar \Omega \sum_{M,m=-\infty}^{\infty} \sum_{\mu=\pm 1} c_M e^{i \underbrace{\frac{2\pi}{T}(M+Km+L\mu)t}_N} \hat{J}_m \hat{A}_m. \quad (\text{A3})$$

2. Second Order

The second order of the Magnus expansion Eq. (26) with the resonance integral Eq. (27) and the beat-note indices

$$N_j = M_j + m_j K + \mu_j L \quad (\text{A4})$$

can be simplified further if the two-photon resonance condition

$$\begin{aligned} M_1 + m_1 K + \mu_1 L + M_2 + m_2 K + \mu_2 L &= 0 \\ \iff (M_1 = -M_2) \wedge (m_1 = -m_2) \wedge (\mu_1 = -\mu_2) \end{aligned} \quad (\text{A5})$$

is satisfied (see Appendix C).

The general form of the form factors Section III B is

$$\begin{aligned} d_{x,y}^{(n)} &= \mp \frac{\Omega^2 T^2}{2\pi} e^{-\eta^2} \sum_{m \text{ even, odd}} \sum_{M\mu} \frac{|c_M|^2}{M + mK + \mu L} (-\eta^2)^{|m|} \\ &\quad \left(L_{\min(n,n-m)}^{[m]}(\eta^2) \right)^2 \frac{\min(n, n-m)!}{\max(n, n-m)!}. \end{aligned}$$

Appendix B: Choice of fidelity

The Bell state fidelity, for an initial state ρ_0 and a target qubit state $|\psi_{\text{target}}\rangle$,

$$\mathcal{F}_{\text{Bell}} = \langle \psi_{\text{target}} | \text{Tr}_{\text{motion}} \{ \hat{U}(T) \rho_0 \hat{U}^\dagger(T) \} | \psi_{\text{target}} \rangle \quad (\text{B1})$$

is often used in experiments [10, 11, 24, 30, 53, 57, 58] and therefore allows for straightforward comparison. The main advantage of the metric is that it is a good measure of (maximally) achievable entanglement, while at the same time ignoring certain relative phases that can otherwise be corrected, e.g., with single qubit gates. Also, the Bell state fidelity allows for straightforward calculation of the fidelity for Hamiltonians of the form

$$\sum_n \sum_{j=x,y,z} \left(d_j^{(n)} \hat{J}_j^2 \right) \otimes |n\rangle\langle n|. \quad (\text{B2})$$

Since $\hat{J}_x^2$ and $\hat{J}_y^2$ are qubit entangling operators, this is an entangling operation which does not affect the state of the motional mode. It yields for an initial qubit state $|\psi_0\rangle = |00\rangle$, an entangled qubit target state $|\psi_t\rangle = \frac{1}{\sqrt{2}}(|00\rangle + e^{i\phi}|11\rangle)$, and an initial motional mode state $\sum_{n=0}^{\infty} P_n |n\rangle\langle n|$, where P_n is the initial probability for the motional mode to be in state $|n\rangle$, the Bell state fidelity

$$\mathcal{F}_{\text{Bell}} = \frac{1}{2} \left(1 - \sum_n P_n \sin(\phi) \sin(d_x^{(n)} - d_y^{(n)}) \right). \quad (\text{B3})$$

Hence the optimal fidelity $\mathcal{F}_{\text{Bell}} = 1$ for a target state $|\psi_t\rangle = \frac{1}{\sqrt{2}}(|00\rangle - i|11\rangle)$, with rotation angle $\phi = -\frac{\pi}{2}$, is obtained if

$$\sum_n P_n \sin(d_x^{(n)} - d_y^{(n)}) = 1 \quad (\text{B4})$$

is fulfilled. This equation can be solved numerically, but to get an analytical solution, further approximations must be made. Assuming P_n decays quickly with n then one can, e.g., assume a thermal distribution $P_n = \frac{(\bar{n})^n}{(\bar{n}+1)^{n+1}}$, with small average phonon number $\bar{n} \ll 1$. Then, the terms for $n > 0$ can be neglected, and the condition for optimal fidelity becomes $\sin(d_x^{(0)} - d_y^{(0)}) = 1$, and thus

$$d_x^{(0)} - d_y^{(0)} = \frac{\pi}{2}. \quad (\text{B5})$$

This equation is valid for the leading terms of the effective Hamiltonian for the Mølmer-Sørensen gate. However, the Bell state fidelity requires fixing specific initial and target states.

Another measure which allows quantification of the fidelity in quantum computing is the average (gate-overlap) fidelity

$$\mathcal{F}_{\text{av}} = \frac{1}{4} \left| \text{Tr}_{\text{qubits}} \left\{ \sum_{n=0}^{\infty} P_n \langle n | \hat{U}(T) \hat{U}_{\text{target}}^\dagger | n \rangle \right\} \right|, \quad (\text{B6})$$

with $\hat{U}_{\text{target}} = \exp(i\varphi \hat{J}_y^2)$, $\varphi = \frac{\pi}{2}$. It yields nearly the same results as the Bell fidelity if the trace over the motional modes is weighted by the Boltzmann distribution, $P_n = \frac{\bar{n}^n}{(\bar{n}+1)^{n+1}}$ ensuring that performance is measured accordingly for more likely initial states.

The average fidelity is a more standard measure for quantum information settings [34], because it measures how close a gate is to the target gate, which is useful for implementation of quantum algorithms.

In order to calculate Ω_{LD} and Ω_4 , the average fidelity can be used directly.

Comparison of the target propagator and the effective Hamiltonian Eq. (B2) shows that the average fidelity needs small d_x and d_z and additionally

$$d_y = -\frac{\pi}{2}. \quad (\text{B7})$$

To calculate the impact of the $\hat{J}_x^2$ -term, the Bell fidelity is quite handy, since Eq. (B3) is concise expression containing both d_x and d_y . The resulting Ω_2 yields a small correction to Ω_{LD} .

Although the Bell state fidelity requires specification of the involved states and the average fidelity requires a specific target gate, the results are nonetheless quite similar: For the parameters used in Fig. 3 at $\Omega = \Omega_2$, the Bell state infidelity is $1 - \mathcal{F}_{\text{Bell}}(\hat{U}_{\text{num}}) = 1.3 \times 10^{-3}$ and the average infidelity $1 - \mathcal{F}_{\text{av}}(\hat{U}_{\text{num}}) = 0.67 \times 10^{-3}$. At $\Omega = \Omega_4$ the infidelities are $1 - \mathcal{F}_{\text{Bell}}(\hat{U}_{\text{num}}) = 4.3 \times 10^{-4}$ and $1 - \mathcal{F}_{\text{av}}(\hat{U}_{\text{num}}) = 2.4 \times 10^{-4}$ for the Bell state infidelity and the average infidelity, respectively. A comparison over the range of amplitudes is shown in Fig. 5.

Beyond this discussion, the error budget terms presented in Table I provide the precise Hamiltonian description and do not hinge on a particular fidelity metric. As such, they are relevant to any particular cost function that is used.

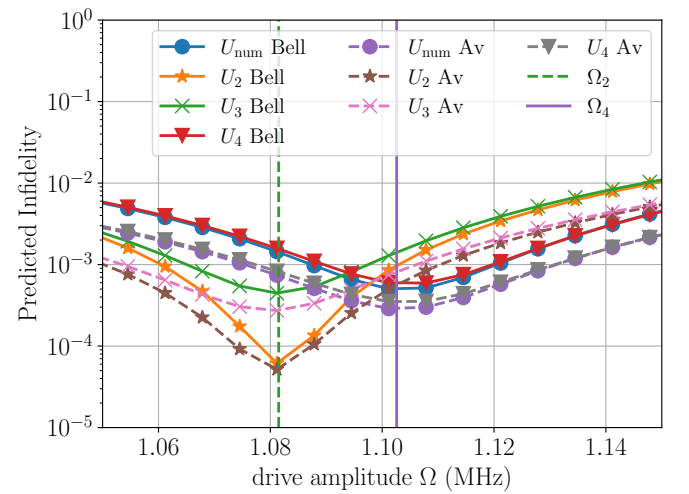


FIG. 5: Comparison of the results for average and Bell state infidelity, using the same parameters as in Fig. 3

Appendix C: Conditions on the variables for the analytical terms

The terms in Table I assume that the detuning avoids certain sideband ranges, which is justified as follows. In order to simplify the expression for $\hat{Z}_2$, Eq. (26), we have assumed in Eq. (28) that

$$N_1 + N_2 = 0 \iff (m_1 + m_2 = 0) \wedge (\mu_1 + \mu_2 = 0) \quad (C1)$$

To drive a single motional mode the laser detuning must be close to the motional mode frequency $K - L \ll K$. If $\mu_1 + \mu_2 \neq 0$, then the resonance condition $N_1 + N_2 = 0$ can be rewritten as

$$1 + \frac{m_1 + m_2}{\mu_1 + \mu_2} = \frac{K - L}{K} \ll 1. \quad (C2)$$

Checking the condition for the allowed values of $\mu_1 + \mu_2$ yields:

1. $1 - \frac{m_1 + m_2}{2} \ll 1$ for $\nu_1 + \nu_2 = -2$ This can only be fulfilled for $m_1 + m_2 = 1$ and yields the resonance condition $K = 2L$.
2. $1 - (m_1 + m_2) \ll 1$ does not yield a result for integer m_1, m_2 .
3. Nor does $1 + (m_1 + m_2) \ll 1$.
4. $1 + \frac{m_1 + m_2}{2} \ll 1$ can be fulfilled for $m_1 + m_2 = -1$ and yields the resonance condition $K = 2L$.

Thus, as long as $K - L \ll K$ and specifically $K \neq 2L$, condition Eq. (28) is valid.

One can make similar assumptions for the other orders of the Magnus expansion. So, for simplicity, to calculate the terms in Table I, we excluded at each k -th order of the Magnus expansion terms where $jK = lL$ if one of the following conditions on $\frac{j-l}{l} = \frac{K-L}{K}$ is true:

1. $\frac{l-j}{l} \leq 0$, which is false, because by construction $K > 0$ and $K - L > 0$.
2. $\frac{l-j}{l} \geq \frac{1}{km_{\max}}$, where $m_{\max}$ is the maximum value of m which is included in the calculation of the Magnus expansion (for our calculations $m_{\max} = 3$).

In most cases where $\frac{K-L}{K} \ll 1$, the second condition is wrong as well. For small K and larger $K - L$ (e.g. $K = 28$ and $K - L = 3$), one has to make sure that there is no $j \leq km_{\max}$ and $|l| \leq k$ such that $jK = lL$ (which is fine for $K = 28$ and $K - L = 3$).

Appendix D: Full expressions for Magnus expansion of the shaped pulse

The full expressions for the Magnus expansion for a $\sin^2$ -pulse are the following: For the 2nd order of the Magnus expansion, Eq. (45), the coefficients are

$$p_y = 3(K^2 - L^2)^2 - 4(5K^2 + 3L^2 - 8), \quad (D1)$$

$$q_y = 8(K^2 - L^2) \left((K - L)^2 - 4 \right) \left((K + L)^2 - 4 \right) \quad (D2)$$

for the first sideband, and

$$p_x = 8(6K^4 - 3K^2L^2 - 10K^2 + 4) + 3(L^4 - 4L^2), \quad (D3)$$

$$q_x = 8(4K^2 - L) \left((2K - L)^2 - 4 \right) \left((2K + L)^2 - 4 \right) \quad (D4)$$

for the second sideband.

The optimal drive amplitude (taking only the leading Lamb-Dicke order and the first sideband into account) is shown in Eq. (46).

The coefficients for the 3rd order of the Magnus expansion, Eq. (47), are

$$p_3 = (K^2 + 3L^2 - 4) \left(3(K^2 - L^2)^2 - 4(5K^2 + 3L^2 - 8) \right), \quad (D5)$$

$$q_3 = 8(K^2 - L^2)^2 \left((K - L)^2 - 4 \right)^2 \left((K + L)^2 - 4 \right)^2. \quad (D6)$$